Continue: Differentiation.

[3] $y = (\sinh^{-1} x)^2$

$$y' = 2\sinh^{-1} x\left(\frac{1}{\sqrt{1+x^2}}\right) = 2\frac{\sinh^{-1} x}{\sqrt{1+x^2}} \qquad (*\sqrt{1+x^2})$$

$$\sqrt{1+x^2}\,y' = 2\sinh^{-1} x$$

Differentiate both sides with respect to x

$$\frac{1}{2\sqrt{1+x^2}}(2x)y' + \sqrt{1+x^2}\,y'' = 2\frac{1}{\sqrt{1+x^2}} \qquad (*\sqrt{1+x^2})$$

$$(1+x^2)y'' + xy' = 2$$

[4] $y = (\sin^{-1} x)^2$

$$y' = 2\sin^{-1} x\left(\frac{1}{\sqrt{1-x^2}}\right) = 2\frac{\sin^{-1} x}{\sqrt{1-x^2}} \qquad (*\sqrt{1-x^2})$$

$$\sqrt{1-x^2}\,y' = 2\sin^{-1} x$$

Differentiate both sides with respect to x

$$\frac{1}{2\sqrt{1-x^2}}(-2x)y' + \sqrt{1-x^2}\,y'' = 2\frac{1}{\sqrt{1-x^2}} \qquad (*\sqrt{1-x^2})$$

$$-xy' + (1-x^2)y'' = 2$$

Differentiate both sides with respect to x

$$-y' - xy'' + (-2x)y'' + (1-x^2)y''' = 0$$

$$(1-x^2)y''' - 3xy'' - y' = 0$$

[5] (a) $y=[\cos x]^3-12\sin(x^2)$	$\Rightarrow y'=3\cos^2 x.(-\sin x)-12\cos(x^2).(2x)$
(b) $y=\tan(8x)+2[\sec(12x)]^3$	$\Rightarrow y'=\sec^2(8x).(8)+6\sec^2(12x).\sec(12x).\tan(12x).(12)$
(c) $y=4^{(7x^2)}[\tan(6x)]^2$	$\Rightarrow y'=4^{(7x^2)}(\ln 4).(14x)\tan^2(6x)+4^{(7x^2)}(2\tan(6x))(\sec^2(6x))(6)$
(d) $y=[\cot(7x)]^3+[\csc(6x)]^4$	$\Rightarrow y'=3\cot^2(7x).(-\csc^2(7x)).(7)+4\csc^3(6x).(-\csc(6x).\cot(6x)).(6)$
(e) $y=\cos(2x^2)-[\cot(9x)]^3$	$\Rightarrow y'=-\sin(2x^2).(4x)-3\cot^2(9x).(-\csc^2(9x)).(9)$
(f) $y=(x-4).[\cos(9x)]^2$	$\Rightarrow y'=\cos^2(9x)+(x-4).2\cos(9x).(-\sin(9x)).(9)$

(g) $y=\dfrac{1-\tan x}{1+\tan x}$

$$y'=\frac{[(1+\tan x).(-\sec^2 x)]-[(1-\tan x).(\sec^2 x)]}{(1+\tan x)^2}=\frac{\left(-\sec^2 x-\tan x\sec^2 x\right)-\left(\sec^2 x-\sec^2 x\tan x\right)}{(1+\tan x)^2}$$

$$y'=\frac{-2\sec^2 x}{(1+\tan x)^2}$$

(h) $y=(x+\cot x)^{\frac{5}{2}}$	$\Rightarrow y'=\dfrac{5}{2}(x+\cot x)^{\frac{3}{2}}.(1-\csc^2 x)$
(i) $y=[\tan(x^2)]^5$	$\Rightarrow y'=5(\tan(x^2))^4.(\sec^2(x^2)).(2x)$
(j) $y=\sqrt{\sin\sqrt{x}}$	$\Rightarrow y'=\dfrac{1}{2\sqrt{\sin\sqrt{x}}}.(\cos\sqrt{x}).\dfrac{1}{2\sqrt{x}}$
(k) $y=[\cot x]^2+\cot x^2$	$\Rightarrow y'=2\cot x.(-\csc^2 x)+(-\csc^2 x^2).(2x)$
(l) $y=\sin\left[\tan\sqrt{\sin x}\right]$	$\Rightarrow y'=\cos\left[\tan\sqrt{\sin x}\right]\left[\sec^2\sqrt{\sin x}\right]\left(\dfrac{1}{2\sqrt{\sin x}}\right).(\cos x)$
(m) $y=\cos(\sqrt{1+x^2})$	$\Rightarrow y'=-\sin(\sqrt{1+x^2}).\left(\dfrac{1}{2\sqrt{1+x^2}}\right).(2x)$
(n) $y=\cos^{-1}(\sin^{-1} x)$	$\Rightarrow y'=\dfrac{-1}{\sqrt{1-(\sin^{-1}x)^2}}.\dfrac{1}{\sqrt{1-x^2}}$
(o) $y=\sin^{-1}(3x+1)$	$\Rightarrow y'=\dfrac{1}{\sqrt{1-(3x+1)^2}}.(3)$
(p) $y=\ln x.\sec^{-1}(3x^2)$	$\Rightarrow y'=\left(\dfrac{1}{x}\right)\sec^{-1}(3x^2)+(\ln x)\left(\dfrac{1}{(3x^2)\sqrt{(3x^2)^2-1}}\right)(6x)$
(q) $y=\sin^{-1}(\tan^{-1}x)$	$\Rightarrow y'=\dfrac{1}{\sqrt{1-(\tan^{-1}x)^2}}.\dfrac{1}{1+x^2}$
(r) $y=\cot^{-1}\left(\dfrac{1+x}{1-x}\right)$	$\Rightarrow y'=\dfrac{-1}{1+\left(\dfrac{1+x}{1-x}\right)^2}.\dfrac{(1-x)(1)-(1+x)(-1)}{(1-x)^2}$

(s) $y=\dfrac{x+\sec^{-1}x}{x^2+\tan^{-1}x}$	$\Rightarrow y'=\dfrac{\left[(x^2+\tan^{-1}x).\left(1+\dfrac{1}{x\sqrt{x^2-1}}\right)\right]-\left[(x+\sec^{-1}x).\left(2x+\dfrac{1}{1+x^2}\right)\right]}{(x^2+\tan^{-1}x)^2}$
(t) $y=\dfrac{1}{x^2}+\sin^{-1}(2x-1)$	$\Rightarrow y'=\dfrac{-2}{x^3}+\dfrac{1}{\sqrt{1-(2x-1)^2}}.(2)$
(u) $y=\tan^{-1}(x^3)$	$\Rightarrow y'=\dfrac{1}{1+(x^3)^2}.(3x^2)$
(v) $y=(\cot^{-1}x)^3$	$\Rightarrow y'=3(\cot^{-1}x)^2.\dfrac{-1}{1+x^2}$
(w) $y=\sec^{-1}\sqrt{1+x^2}$	$\Rightarrow y'=\dfrac{1}{\sqrt{1+x^2}.\sqrt{(1+x^2)-1}}.\dfrac{1}{2\sqrt{1+x^2}}.(2x)$
(x) $y=\tan^{-1}\left(\dfrac{x}{a}\right)$	$\Rightarrow y'=\dfrac{1}{1+\left(\dfrac{x}{a}\right)^2}.\dfrac{1}{a}$
(y) $y=\sqrt{\csc^{-1}x}$	$\Rightarrow y'=\dfrac{1}{2\sqrt{\csc^{-1}x}}.\dfrac{-1}{x\sqrt{x^2-1}}$
(z) $y=(\tan^{-1}x)^{-1}+\tan^{-1}(\sin x)$	$\Rightarrow y'=-(\tan^{-1}x)^{-2}.\dfrac{1}{1+x^2}+\dfrac{1}{1+(\sin x)^2}.\cos x$
(i) $y=(x^2+1)+\sin^{-1}\left(\dfrac{x}{a}\right)$	$\Rightarrow y'=2x+\dfrac{1}{\sqrt{1-\left(\dfrac{x}{a}\right)^2}}.\dfrac{1}{a}$
(ii) $y=\sqrt{x^2-1}+\sin^{-1}x$	$\Rightarrow y'=\dfrac{1}{2\sqrt{x^2-1}}.(2x)+\dfrac{1}{\sqrt{1-x^2}}$

(iii) $y=\cos(\sqrt{1+\sin^2 x})-\csc^{-1}(3x)$

$$y'=\left[-\sin(\sqrt{1+\sin^2 x}).\left(\frac{1}{2\sqrt{1+\sin^2 x}}\right).(2\sin x.\cos x)\right]-\left[\frac{-1}{3x\sqrt{(3x)^2-1}}.(3)\right]$$

(iv) $y=[\sec^2(3x)+\cos^{-1}x]^4$

$$y'=4[\sec^2(3x)+\cos^{-1}x]^3.\left[[2\sec(3x).\sec(3x)\tan(3x).(3)]+\left(\frac{-1}{\sqrt{1-x^2}}\right)\right]$$

(v) $y=\sin^{-1}\left(\dfrac{x-1}{x+1}\right)$	$\Rightarrow y'=\dfrac{1}{\sqrt{1-\left(\dfrac{x-1}{x+1}\right)^2}}.\dfrac{(x+1).(1)-(x-1).(1)}{(x+1)^2}$
(vi) $y=\tan^{-1}(\sin^{-1}\sqrt{x})$	$\Rightarrow y'=\dfrac{1}{1+(\sin^{-1}\sqrt{x})^2}.\dfrac{1}{\sqrt{1-(\sqrt{x})^2}}.\dfrac{1}{2\sqrt{x}}$
(vii) $y=[\cosh(\sqrt{2x+1})]^{-4}$	$\Rightarrow y'=-4\left[\cosh(\sqrt{2x+1})\right]^{-5}.\left[\sinh(\sqrt{2x+1})\right]\left(\dfrac{1}{2\sqrt{2x+1}}\right)(2)$

(viii) $y=(\log_2 x).(\sec h(e^{2x}))$

$$\Rightarrow y'=\left(\frac{1}{x\ln 2}\right)(\sec h(e^{2x}))+(\log_2 x)(-\sec h(e^{2x})\tanh(e^{2x}))(e^{2x})(2)$$

(ix) $y=\tan^2(6x-4)$ $\Rightarrow y'=2\tan(6x-4)\sec^2(6x-4).(6)=12\tan(6x-4)\sec^2(6x-4)$

(x) $y=x^2\csc^4(6x)$

$y'=x^2.4\csc^3(6x).(-).\csc(6x)\cot(6x).(6)+2x\csc^4(6x)$

or $\ln y=2\ln x+4\ln(\csc(6x))$ Differentiate with respect to x

$\frac{1}{y}y'=\frac{2}{x}-\frac{24\csc(6x)\cot(6x)}{\csc(6x)}$ $\Rightarrow y'=y.\left[\frac{2}{x}-\frac{24\csc(6x)\cot(6x)}{\csc(6x)}\right]=y.\left[\frac{2}{x}-24\cot(6x)\right]$

$\Rightarrow y'=x^2\csc^4(6x).\left[\frac{2}{x}-24\cot(6x)\right]$

(xi) $y=\cosh^3 x-12\sin^{-1}x$ $\Rightarrow y'=3\cosh^2 x\sinh x-\frac{12}{\sqrt{1-x^2}}$

(xii) $y=2\sec h^{-1}(12x)$ $\Rightarrow y'=\frac{-24}{12x\sqrt{1-144x^2}}=\frac{-2}{x\sqrt{1-144x^2}}$

(xiii) $y=x^3\coth^2(6x)$

$y'=x^3.2\coth(6x).(-).\csc h^2(6x).(6)+3x^2.\coth^2(6x)$

or $\ln y=3\ln x+2\ln(\coth(6x))$ Differentiate with respect to x

$\frac{1}{y}y'=\frac{3}{x}+\frac{2(-\csc h^2(6x))(6)}{\coth(6x)}$ $\Rightarrow y'=y.\left[\frac{3}{x}+\frac{2(-\csc h^2(6x))(6)}{\coth(6x)}\right]=y.\left[\frac{3}{x}-\frac{12\csc h^2(6x)}{\coth(6x)}\right]$

$\Rightarrow y'=x^3\coth^2(6x).\left[\frac{3}{x}-\frac{12}{\cosh(6x)\sinh(6x)}\right]$

(xiv) $y=(\tanh x)^{-1}$ $\Rightarrow y'=-(\tanh x)^{-2}(\sec h^2 x)$

(xv) $y=\tanh^{-1}(2x)+\csc h^{-1}(2x)$ $\Rightarrow \boldsymbol{y'=\frac{2}{1-4x^2}-\frac{2}{2x\sqrt{4x^2+1}}}$

Some Applications on Differentiation.

I – Logarithmic Differentiation:

Remember that $\ln\left(\frac{ab}{cd}\right) = \ln(a)+\ln(b)-\ln(c)-\ln(d)$

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[6] (a) $\mathbf{y = \frac{\sin^2 x \tanh^4 x}{(x^2+1)^2}}$ take $\ln(.)$ for both sides

$\mathbf{\ln y = 2\ln(\sin x) + 4\ln(\tanh x) - 2\ln(x^2+1)}$ Differentiate with respect to x

$$\mathbf{\frac{1}{y}y' = \frac{2\cos x}{\sin x} + \frac{4\operatorname{sech}^2 x}{\tanh x} - \frac{4x}{x^2+1} \quad \Rightarrow y' = y.\left[\frac{2\cos x}{\sin x} + \frac{4\operatorname{sech}^2 x}{\tanh x} - \frac{4x}{x^2+1}\right]}$$

$$\mathbf{\Rightarrow y' = \frac{\sin^2 x \tanh^4 x}{(x^2+1)^2}.\left[\frac{2\cos x}{\sin x} + \frac{4\operatorname{sech}^2 x}{\tanh x} - \frac{4x}{x^2+1}\right]}$$

(b) $y = \frac{(x-1)^3\sqrt{x-2}}{\sqrt[4]{x-6}}$ take $\ln(.)$ for both sides

$\ln y = 3\ln(x-1) + \frac{1}{2}\ln(x-2) - \frac{1}{4}\ln(x-6)$ Differentiate with respect to x

$$\frac{1}{y}y' = \frac{3}{x-1} + \frac{1}{2(x-2)} - \frac{1}{4(x-6)} \quad \Rightarrow y' = y.\left[\frac{3}{x-1} + \frac{1}{2(x-2)} - \frac{1}{4(x-6)}\right]$$

$$\Rightarrow y' = \frac{(x-1)^3\sqrt{x-2}}{\sqrt[4]{x-6}}.\left[\frac{3}{x-1} + \frac{1}{2(x-2)} - \frac{1}{4(x-6)}\right]$$

(c) $y = \sqrt{x.\sinh^{-1} x.\sqrt{1-e^x}} = \left[x.\sinh^{-1} x.\sqrt{1-e^x}\right]^{\frac{1}{2}}$ take $\ln(.)$ for both sides

$\ln y = \frac{1}{2}\left[\ln x + \ln(\sinh^{-1} x) + \frac{1}{2}\ln(1-e^x)\right]$ Differentiate with respect to x

$$\frac{1}{y}y' = \frac{1}{2x} + \frac{1}{2\sinh^{-1} x}\frac{1}{\sqrt{1+x^2}} + \frac{1}{4(1-e^x)}(-e^x) \quad \Rightarrow y' = y\left[\frac{1}{2x} + \frac{1}{2\sinh^{-1} x}\frac{1}{\sqrt{1+x^2}} + \frac{1}{4(1-e^x)}(-e^x)\right]$$

$$\Rightarrow y' = \sqrt{x.\sinh^{-1} x.\sqrt{1-e^x}}\left[\frac{1}{2x} + \frac{1}{2\sinh^{-1} x}\frac{1}{\sqrt{1+x^2}} + \frac{1}{4(1-e^x)}(-e^x)\right]$$

(d) $y = (\sin x)^x$ take $\ln(.)$ for both sides

$\ln y = x\ln(\sin x)$ Differentiate with respect to x

$$\frac{1}{y}y' = \ln(\sin x) + \frac{x\cos x}{\sin x} \quad \Rightarrow y' = y.\left[\ln(\sin x) + \frac{x\cos x}{\sin x}\right]$$

$$\Rightarrow y' = (\sin x)^x.\left[\ln(\sin x) + \frac{x\cos x}{\sin x}\right]$$

(e) $y = x^{\ln x}$ take $\ln(.)$ for both sides

$\ln y = \ln x . \ln x = (\ln x)^2$ Differentiate with respect to x

$$\frac{1}{y} y' = \frac{2\ln x}{x} \qquad \Rightarrow y' = y\left(\frac{2\ln x}{x}\right) = x^{\ln x} . \left(\frac{2\ln x}{x}\right)$$

(f) $y = (\ln x)^{\cos x}$ take $\ln(.)$ for both sides

$\ln y = \cos x . \ln(\ln x)$ Differentiate with respect to x

$$\frac{1}{y} y' = -\sin x . \ln(\ln x) + \frac{\cos x}{x \ln x} \qquad \Rightarrow y' = y . \left[-\sin x . \ln(\ln x) + \frac{\cos x}{x \ln x}\right]$$

$$\Rightarrow y' = (\ln x)^{\cos x} . \left[-\sin x . \ln(\ln x) + \frac{\cos x}{x \ln x}\right]$$

(g) $y = (\sinh x)^{\frac{1}{x}}$ take $\ln(.)$ for both sides

$\ln y = \frac{1}{x}(\ln(\sinh x))$ Differentiate with respect to x

$$\frac{1}{y} y' = \frac{\cosh x}{x \sinh x} - \frac{\ln(\sinh x)}{x^2} \qquad \Rightarrow y' = y . \left[\frac{\cosh x}{x \sinh x} - \frac{\ln(\sinh x)}{x^2}\right]$$

$$\Rightarrow y' = (\sinh x)^{\frac{1}{x}} . \left[\frac{\cosh x}{x \sinh x} - \frac{\ln(\sinh x)}{x^2}\right]$$

(h) $y = (\operatorname{csc} h^{-1} x)^2 + (\coth x)^{\sqrt{x}}$

Let $y = u + v$ where $u = (\operatorname{csc} h^{-1} x)^2$ and $v = (\coth x)^{\sqrt{x}}$

$y' = u' + v'$

$$\boxed{u' = 2(\operatorname{csc} h^{-1} x)\left(\frac{-1}{|x| \sqrt{1+x^2}}\right)}$$

$v = (\coth x)^{\sqrt{x}}$ take $\ln(.)$ for both sides

$\ln v = \sqrt{x} \ln(\coth x)$ Differentiate with respect to x

$$\frac{1}{v} v' = \frac{1}{2\sqrt{x}} \ln(\coth x) + \sqrt{x} \frac{1}{\coth x}(-\operatorname{csc} h^2 x) \quad \Rightarrow v' = v\left[\frac{1}{2\sqrt{x}} \ln(\coth x) + \sqrt{x} \frac{1}{\coth x}(-\operatorname{csc} h^2 x)\right]$$

$$\Rightarrow \boxed{v' = (\coth x)^{\sqrt{x}}\left[\frac{1}{2\sqrt{x}} \ln(\coth x) + \sqrt{x} \frac{1}{\coth x}(-\operatorname{csc} h^2 x)\right]}$$

$$\Rightarrow y' = u' + v' = 2(\operatorname{csc} h^{-1} x)\left(\frac{-1}{|x| \sqrt{1+x^2}}\right) + (\coth x)^{\sqrt{x}}\left[\frac{1}{2\sqrt{x}} \ln(\coth x) + \sqrt{x} \frac{1}{\coth x}(-\operatorname{csc} h^2 x)\right]$$

II – Implicit Differentiation:

[7] (a) $x^2y+3xy^2-x=3$ Differentiate with respect to x

$$\mathbf{2xy+x^2y'+3y^2+3x.(2y)y'-1=0}$$

$$\mathbf{(x^2+6xy)y'+(2xy+3y^2-1)=0}$$

$$\mathbf{y'=\frac{1-2xy-3y^2}{x^2+6xy}}$$

(b) $\cos(xy)=y$ Differentiate with respect to x

$$-\sin(xy).[y+xy']=y'$$

$$[-x\sin(xy)-1]y'=y\sin(xy)$$

$$y'=\frac{-y\sin(xy)}{x\sin(xy)+1}$$

(c) $\sin(x^2y^2)=x$ Differentiate with respect to x

$$\cos(x^2y^2).[2xy^2+2x^2yy']=1$$

$$y'=\frac{1-2xy^2\cos(x^2y^2)}{2x^2y\cos(x^2y^2)}$$

(d) $\dfrac{xy^3}{1+\sec y}=1+y^4 \quad \Leftrightarrow xy^3=(1+y^4).(1+\sec y)$ Differentiate with respect to x

$$y^3+3xy^2y'=(4y^3y').(1+\sec(y))+(1+y^4).(\sec(y)\tan(y)y')$$

$$[3xy^2-4y^3(1+\sec(y))-(1+y^4)(\sec(y)\tan(y))].y'=-y^3$$

$$y'=\frac{-y^3}{[3xy^2-4y^3(1+\sec(y))-(1+y^4)(\sec(y)\tan(y))]}$$

(e) $x^2y^2-y\ln x=10$ Differentiate with respect to x

$$2xy^2+x^2(2yy')-\left(y'\ln x+y\frac{1}{x}\right)=0 \qquad (2x^2y-\ln x)y'+2xy^2-\frac{y}{x}=0$$

$$y'=\frac{\frac{y}{x}-2xy^2}{2x^2y-\ln x}$$

(f) $\tan(\sqrt{x})-y.\sec(\sqrt{y})=1$ Differentiate with respect to x

$$\sec^2(\sqrt{x})\frac{1}{2\sqrt{x}}-\left(y'.\sec(\sqrt{y})+y\sec(\sqrt{y})\tan(\sqrt{y})\frac{1}{2\sqrt{y}}y'\right)=0$$

$$\mathbf{y'=\frac{\sec^2(\sqrt{x})\frac{1}{2\sqrt{x}}}{\sec(\sqrt{y})+y\sec(\sqrt{y})\tan(\sqrt{y})\frac{1}{2\sqrt{y}}}}$$

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